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Alg 2 Trig/Honors

Chp 7.4 & 7.5 : Basic Trigonometric Equations & More Trigonometric Equations

A trigonometric equation is an equation that contains trigonometric functions.

I. Basic Trigonometric Equations

Solving any trigonometric equation always reduces to solving a basic trigonometric equation, that is, an equation of the form $T(\theta) = c$, where T is a trigonometric function and c is a constant.

Solving Basic Trigonometric Equations

1. Find the solutions in one period
2. Find all solutions of the equation by adding integer multiples of the period to the solutions in part 1.

II. Solving Trigonometric Equations by Factoring

Factoring is one of the most useful techniques for solving equations, including trigonometric equations. The idea is to move all terms to one side of the equation, factor, and then use the Zero-Product Property to solve.

State the Zero-Product Property.

If $AB = 0$, then $A = 0$ or $B = 0$.

III. Solving Trigonometric Equations by Using Identities and equations with Trigonometric Functions of Multiple Angles

When solving trigonometric equations that involve functions of multiples of angles, first solve for the multiple of the angle, then divide to solve for the angle.

Pythagorean identity.

Using a graphical solution.

Using an algebraic solution

Double-Angle Formula or Half Angle Formulas

Addition or Subtraction Formulas

Sum-To-Product or Product-to-Sum Formulas

Example 1: Solve the given equation, and list six specific solutions.

a. $\sin \theta = -\frac{\sqrt{3}}{2}, -\frac{\pi}{3}$

$$\sin \theta = -\frac{\sqrt{3}}{2}$$

$$\theta = \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$$

$$Q4 \rightarrow \theta_1 = -\frac{\pi}{3} + 2\pi n$$

$$Q3 \rightarrow \theta_2 = -\frac{2\pi}{3} + 2\pi n$$

$$Q4 = \frac{5\pi}{3} + 2\pi n$$

$$Q3 = \frac{4\pi}{3} + 2\pi n$$

b. $\tan \theta = -10$

$$\theta = \tan^{-1}(-10)$$

$$= \underbrace{-1.471127674}_{\text{5th x}} + \pi n$$

$$n=0, \theta = -1.4711$$

$$n=1, \theta = 1.670$$

$$n=2, \theta = 4.8120$$

$$n=3, \theta = 7.9536$$

$$n=4, \theta = 11.0452$$

$$n=5, \theta = 14.2368$$

$$n=0, \theta = -\frac{\pi}{3}, -\frac{2\pi}{n}$$

$$n=1, \theta = \frac{5\pi}{3}, \frac{4\pi}{3}$$

$$n=2, \theta = \frac{11\pi}{3}, \frac{10\pi}{3}$$

$$\begin{array}{l} \theta = \theta - \theta_{\text{ref}} \quad \theta_{\text{ref}} = 0 \\ \theta = \pi + \theta_{\text{ref}} \quad \theta = 2\pi - \theta_{\text{ref}} \end{array}$$

Example 2: Solve the given equation

a. $3\sin^2\theta - 1 = 0, u = \sin\theta$

$$3u^2 - 1 = 0$$

$$3u^2 = 1$$

$$u^2 = \frac{1}{3}$$

$$u = \pm \frac{\sqrt{3}}{3}$$

$$\sin\theta = \pm \frac{\sqrt{3}}{3}$$

$$\sin\theta = \frac{\sqrt{3}}{3}$$

$$\theta_1 = \sin^{-1}\left(\frac{\sqrt{3}}{3}\right)$$

$$\sin\theta = -\frac{\sqrt{3}}{3}$$

$$\theta_2 = \sin^{-1}\left(-\frac{\sqrt{3}}{3}\right)$$

$$\theta_1 = \pm 0.6154 + 2\pi n$$

$$\theta_2 = \pm 2.5261 + 2\pi n$$

b. $2\sin^2\theta + 5\sin\theta - 12 = 0$

$$u = \sin\theta$$

$$2u^2 + 5u - 12 = 0$$

$$(2u^2 + 8u) + (-3u) = 0$$

$$2u(u+4) + (-3)(u+4) = 0$$

$$(2u-3)(u+4) = 0$$

$$2u-3=0 \text{ or } u+4=0$$

$$u = \frac{3}{2} \text{ or } u = -4$$

$$\sin\theta = \frac{3}{2} \text{ or } \sin\theta = -4$$

No solution

c. $\sec\theta(2\cos\theta - \sqrt{2}) = 0$

f. $(\tan\theta - 2)(16\sin^2\theta - 1) = 0$

$$\tan\theta - 2 = 0$$

$$\tan\theta = 2$$

$$\theta = \tan^{-1}(2)$$

$$\theta = 1.1071 + \pi n$$

$$16\sin^2\theta - 1 = 0$$

$$\sin^2\theta = \frac{1}{16}$$

$$\sin\theta = \pm \frac{1}{4}$$

$$\theta = \sin^{-1}\left(\pm \frac{1}{4}\right)$$

$$\theta_1 = \pm 0.2526 + 2\pi n$$

$$\theta_2 = \pm 2.8889 + 2\pi n$$

g. $3\tan^3\theta = \tan\theta$

$$3\tan^3\theta - \tan\theta = 0$$

$$\tan\theta(3\tan^2\theta - 1) = 0$$

$$\tan\theta = 0, \quad 3\tan^2\theta - 1 = 0$$

$$\theta = \tan^{-1}(0)$$

$$\theta = 0 + \pi n$$

$$\theta = \pi n$$

$$\tan^2\theta = \frac{1}{3} \rightarrow \tan\theta = \pm \frac{\sqrt{3}}{3}$$

$$\theta = \tan^{-1}\left(\pm \frac{\sqrt{3}}{3}\right)$$

$$\theta_2 = \frac{\pi}{6} + \pi n$$

$$\theta_3 = -\frac{\pi}{6} + \pi n$$

h. $4\cos\theta\sin\theta + 3\cos\theta = 0$

d. $\sin^2\theta = 4 - 2\cos^2\theta$

i. $\csc^2\theta = \cot\theta + 3 \quad u = \cot\theta$

$$1 + \cot^2\theta = \cot^2\theta + 3$$

$$\cot^2\theta - \cot\theta - 2 = 0$$

$$u^2 - u - 2 = 0$$

$$(u-2)(u+1) = 0$$

$$u = 2 \text{ or } u = -1$$

$$\cot\theta = 2 \text{ or } \cot\theta = -1$$

$$\theta = \cot^{-1}(2)$$

$$= \tan^{-1}\left(\frac{1}{2}\right)$$

$$\theta = 0.4636 + \pi n$$

$$\theta = -\frac{\pi}{4} + \pi n$$

e. $3 \cos 2\theta - 2 \cos^2 \theta = 0$

j. $2 \tan \theta + \sec^2 \theta = 4$

Example 3: Use a graphing device to find the solutions of the equation, rounded to two decimal places.

$$\sin x = x^3$$

1) $y_1 = x^3 - \sin x \rightarrow$ looking for x-ints

2) $y_1 = \sin x$
 $y_2 = x^3$ > looking for intersection pts

$$x \approx -0.93, 0.93, 0$$

Example 4:

- Find all solutions of the equation.
- List the solutions in the interval $[0, 2\pi)$.

a. $2 \sin 3\theta + 1 = 0$

b. $\sec \frac{\theta}{2} = \cos \frac{\theta}{2}$

$$\frac{1}{\cos(\frac{\theta}{2})} = \cos(\frac{\theta}{2})$$

$$1 = \cos^2(\frac{\theta}{2})$$

$$\pm 1 = \cos(\frac{\theta}{2})$$

$$\cos^{-1}(\pm 1) = \frac{\theta}{2}$$

$$\frac{\theta}{2} = 2\pi n$$

$$\frac{\theta}{2} = \pi + 2\pi n$$

$$\begin{aligned} \theta_1 &= 4\pi n \\ \theta_2 &= 2\pi + 4\pi n \end{aligned}$$

$$n=0: \theta=0, 2\pi$$

$$n=1: \theta=4\pi, 6\pi$$

$$S = \{0\}$$

$$c) (\tan 3\theta + 1)^2 = (\sec 3\theta)^2$$

$$\tan^2 3\theta + 2\tan 3\theta + 1 = \sec^2 3\theta$$

$$\cancel{\sec^2 3\theta} + 2\tan 3\theta = \cancel{\sec^2 3\theta}$$

$$2\tan 3\theta = 0$$

$$\tan 3\theta = 0$$

$$3\theta = \tan^{-1}(0)$$

$$3\theta = \pi n$$

$$1) \theta = \frac{\pi}{3} n$$

$$2) n=0: \theta = 0 \quad n=5: \theta = \frac{5\pi}{3}$$

$$n=1: \theta = \frac{\pi}{3} \quad \times$$

$$n=2: \theta = \frac{2\pi}{3}$$

$$n=3: \theta = \pi$$

$$n=4: \theta = \frac{4\pi}{3}$$

check each answer

$$d. 3 \tan^3 \theta - 3 \tan^2 \theta - \tan \theta + 1 = 0$$

$$e. \sec \theta \tan \theta - \cos \theta \cot \theta = \sin \theta$$

$$\frac{1}{\cos \theta} \cdot \tan \theta - \cos \theta \cdot \frac{1}{\tan \theta} = \sin \theta$$

$$\frac{\tan \theta}{\cos \theta} - \frac{\cos \theta}{\tan \theta} = \sin \theta$$

$$\frac{\tan^2 \theta - \cos^2 \theta}{\sin \theta} = \sin \theta$$

$$\tan^2 \theta - \cos^2 \theta = \sin^2 \theta$$

$$\tan^2 \theta = \cos^2 \theta + \sin^2 \theta$$

$$\tan^2 \theta = 1$$

$$\theta = \tan^{-1}(\pm 1)$$

$$\theta = \frac{\pi}{4} + \pi n$$

$$\theta = -\frac{\pi}{4} + \pi n$$

$$n=0: \frac{\pi}{4}, -\frac{\pi}{4}$$

$$n=1: \frac{5\pi}{4}, \frac{3\pi}{4}$$

$$n=2: \frac{9\pi}{4}, \frac{7\pi}{4}$$

$$S = \left\{ \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4} \right\}$$

$$f. \sin 2\theta \cos \theta - \cos 2\theta \sin \theta = \frac{\sqrt{3}}{2}$$

$$\sin(2\theta - \theta) = \frac{\sqrt{3}}{2}$$

$$\sin \theta = \frac{\sqrt{3}}{2}$$

$$\theta = \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$$

$$1) \theta = \frac{\pi}{3} + 2\pi n, \theta = \frac{2\pi}{3} + 2\pi n$$

$$2) n=0: \frac{\pi}{3}, \frac{2\pi}{3}$$

$$g. \sin 5\theta - \sin 3\theta = \cos 4\theta$$

$$2 \cos\left(\frac{5\theta+3\theta}{2}\right) \sin\left(\frac{5\theta-3\theta}{2}\right) = \cos 4\theta$$

$$2 \cos(4\theta) \sin(\theta) = \cos 4\theta$$

$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\theta = \frac{\pi}{6} + 2\pi n$$

$$\theta = \frac{5\pi}{6} + 2\pi n$$

$$n=0: \frac{\pi}{6}, \frac{5\pi}{6}$$

$$S = \left\{ \frac{\pi}{6}, \frac{5\pi}{6} \right\}$$

$$h. \cos \theta - \sin \theta = \sqrt{2} \sin \frac{\theta}{2}$$